

UNMASK Discovering and Causally Verifying Spurious Shortcuts in Text Classifiers

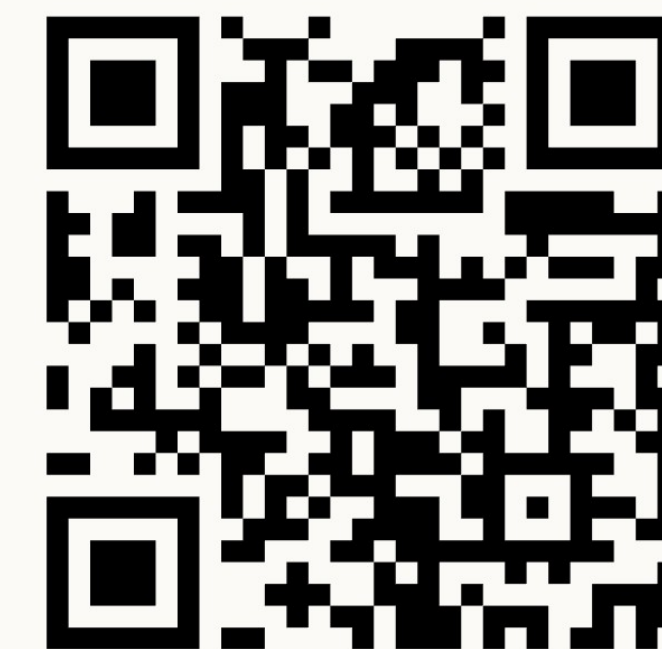
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Problem and Motivation

Text classifiers often rely on **spurious shortcuts**: surface cues that predict the label in the training data but are unrelated to the task.

Example: a BERT toxicity classifier on CivilComments

Comment	P(toxic)	Prediction
You can be fired for being LGBT . [†]	0.91	toxic
You can be fired for being Black .	0.99	toxic
You can be fired for being straight .	0.27	non-toxic
You can be fired for being late .	0.02	non-toxic

[†] A real test comment labeled non-toxic. Other rows change one word. Mean of 3 seeds.

- Finding such shortcuts takes human effort: annotated groups (GroupDRO, DFR) or a hand-built bias model (PoE). Word statistics (PMI) only see single words.
- A cue that correlates with the label is not necessarily one the model relies on.

RQ1 Can we find spurious shortcuts without human annotation?

RQ2 Which of them does a trained model actually rely on?

RQ3 Can they replace human-labeled groups when debiasing the model?

Shortcuts as Code

UNMASK has an LLM write each candidate shortcut as a boolean function over the input text. Three of the functions it found (abridged):

```
any(w in words(h) for w in ["not", "never", "no", ...]) → contradiction
len(words(p) & words(h)) > 2 → entailment
any(w in words(x) for w in ["gay", "LGBTQ", ...]) → toxic
```

p = premise, h = hypothesis (NLI). x = comment (toxicity).

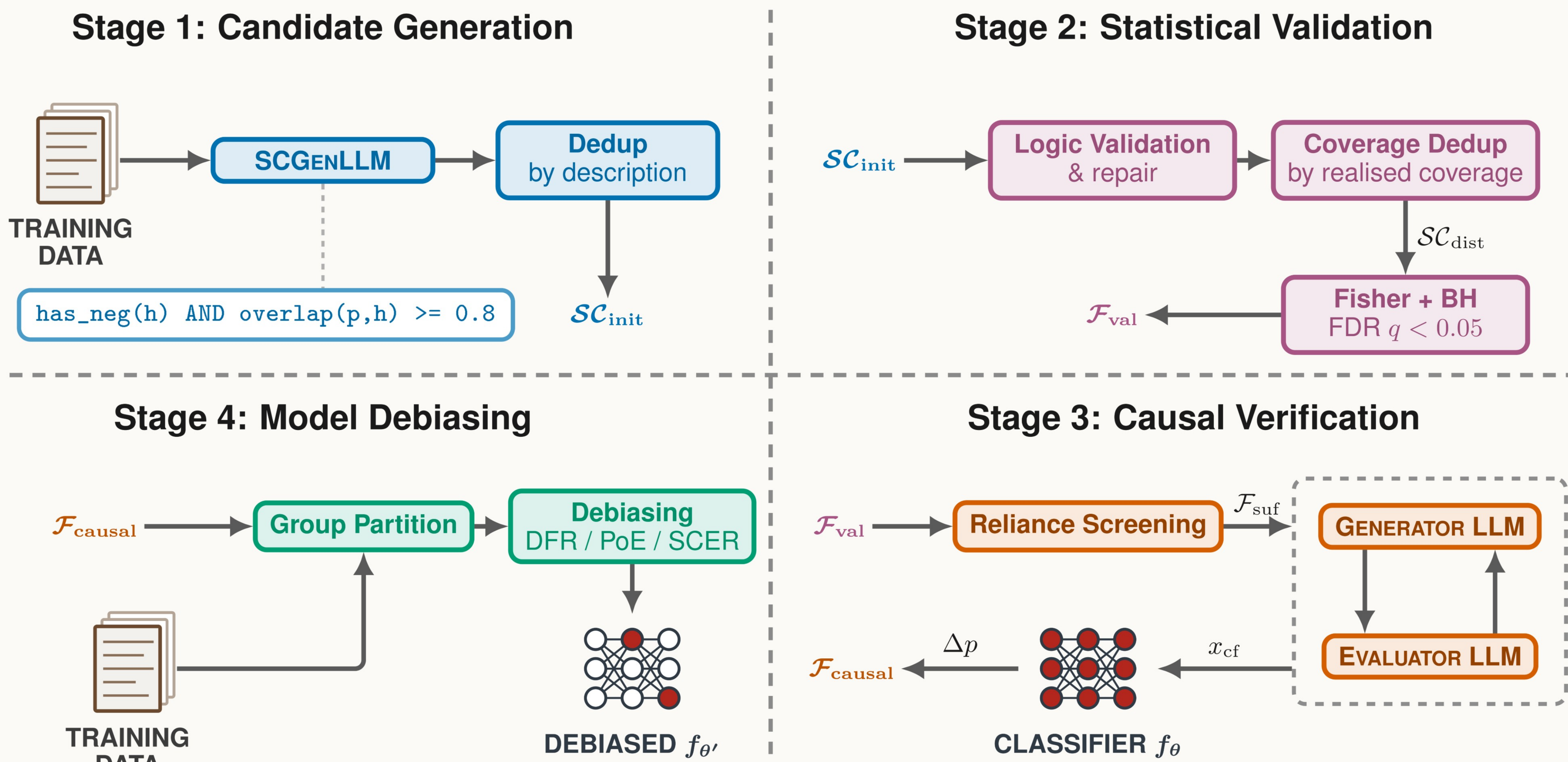
Because each shortcut is code, the same function drives every stage:

- Test**: is it correlated with the label on held-out data?
- Verify**: does the prediction change when it is edited out?
- Debias**: group the training set by it, with no annotation.

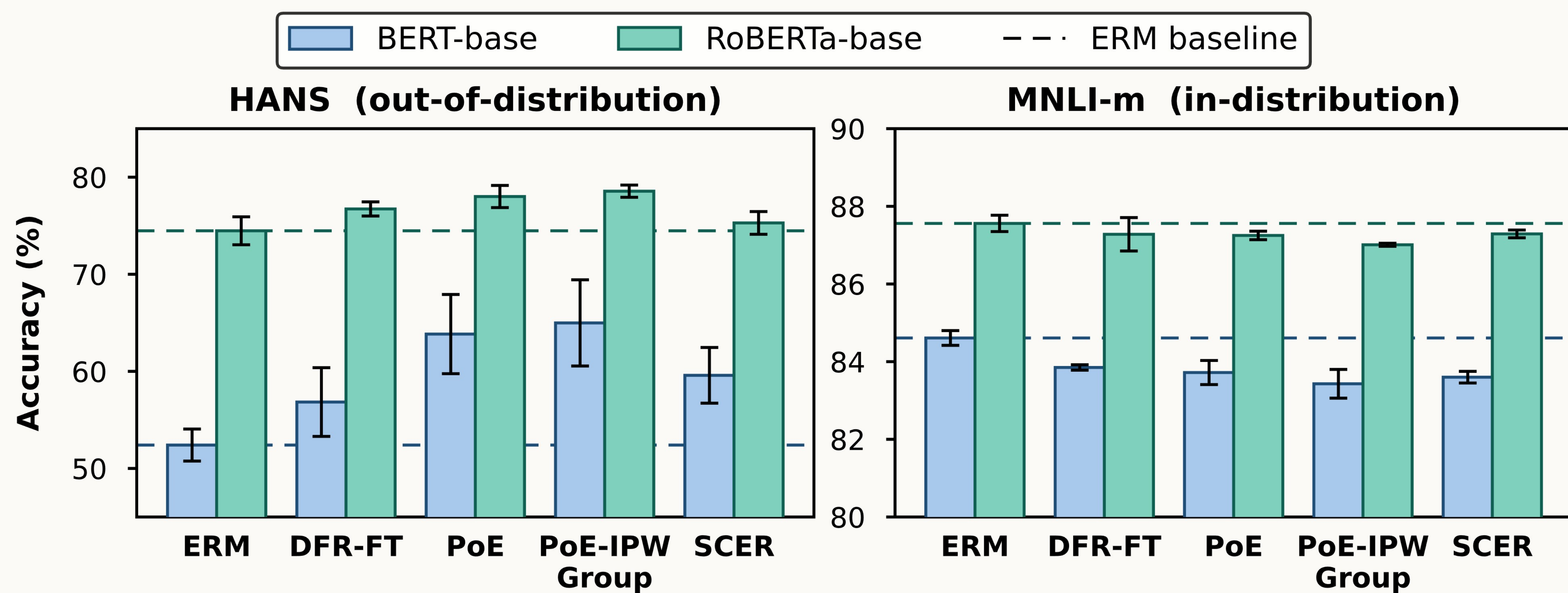
- Correlation is not reliance**. The strongest NLI correlate (OR 10.01) is used by neither BERT nor RoBERTa.

- No labels needed**. On CivilComments the functions recover 6 of the 8 hand-annotated identity groups. Single-word PMI recovers none.

From a Dataset to a Debaised Model

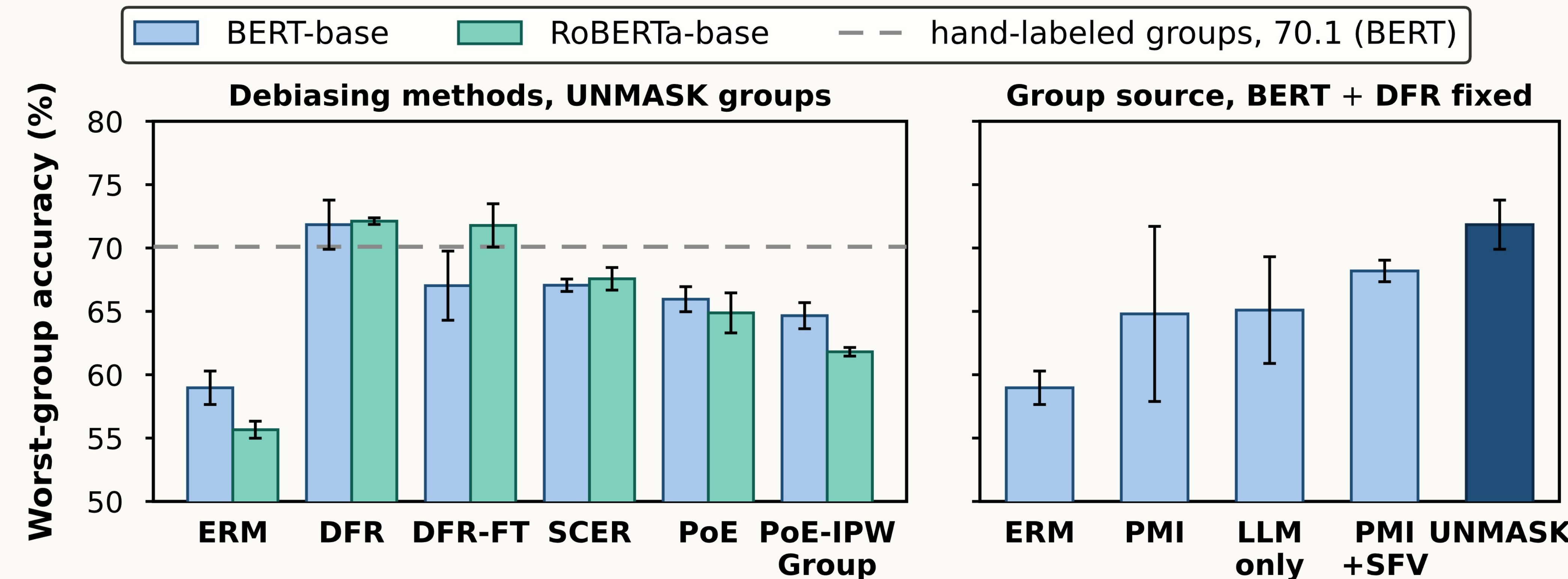


Debiasing NLI While Keeping In-Distribution Accuracy



- HANS accuracy up 12.58 pp on BERT and 4.08 pp on RoBERTa
- MNLI, SNLI and ANLI all stay within 1.3 pp of ERM
- Negative control: on sentiment, verification finds no exploited features (0 of 8 on IMDB, 0 of 5 on SST-2)
- 9 of 10 features verify causally on BERT, 6 of 10 on RoBERTa
- Lexical-overlap features fire on 26–89% of entailment examples, effects of –11 to –15 pp
- DFR-FT keeps the best in-distribution accuracy on BERT while still gaining on HANS

Matching Hand-Labeled Groups on CivilComments



- DFR on UNMASK groups matches hand-labeled DFR (70.1, BERT) without demographic labels: 71.84 on BERT, 72.12 on RoBERTa
- Swap only the group source and UNMASK beats the strongest baseline by 3.65 pp

Spurious Correlations in RewardBench2

Subset	Spurious feature	Correlates with	OR	Prec.
Focus	Response > 300 words	rejected	6.4	0.82
	More than 3 bullet points	rejected	6.1	0.81
Safety	Apologetic language (sorry, apologies)	chosen	25.5	0.89
Math	“let me help you solve ... step by step”	chosen	22.7	0.86
Factuality	Markdown section headers (###)	chosen	2.83	0.47
Ties	Response much longer than the prompt	rejected	2.00	0.84

Chance precision: 0.25 chosen, 0.75 rejected (1 chosen, 3 rejected responses per prompt).

- Reward models are known to favor long, list-heavy answers [1, 2]. In Focus the association runs the other way: responses over 300 words (OR 6.4) or with more than 3 bullets (OR 6.1) go with the **rejected** answer
- Ties has many equally valid answers per prompt [3], yet responses much longer than the prompt still go with the **rejected** answer (OR 2.0)
- Discovery and validation run unmodified on preference data, with no reward model
- 195 significant features on Ties, 163 on Focus, 61 on Safety. None survive on Precise IF.

[1] Singhal et al. A Long Way to Go: Investigating Length Correlations in RLHF. COLM 2024. arxiv.org/abs/2310.03716

[2] Zhang et al. From Lists to Emojis: How Format Bias Affects Model Alignment. ACL 2025. aclanthology.org/2025.acl-long.1308

[3] Malik et al. RewardBench 2: Advancing Reward Model Evaluation. ICLR 2026. arxiv.org/abs/2506.01937